STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 19

NORMAL DISTRIBUTION

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NORMAL DISTRIBUTION

Normal distribution is the most important continuous distribution in the entire field of statistics. It is also called **Gaussian Distribution**. It is widely used in the real world problems as real world phenomena are mostly approximately normally distributed.

EXAMPLES OF NORMAL DISTRIBUTION The following continuous random variables represents approximate normal distribution.

1. Heights and weights of people.
2. Scores on an examination.
3. Amount of milk in a gallon.
4. Time taken to complete a certain job
5. Life of an item (such as a light-bulb or a television set).

DEFINITION AND PDF OF NORMAL DISTRIBUTION

A **normal distribution** is a continuous, symmetric, bell-shaped distribution of a variable.

A random variable X following Normal distribution having mean μ and standard deviation σ is denoted by $X \sim N(\mu, \sigma^2)$. Its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \quad -\infty < x < \infty$$

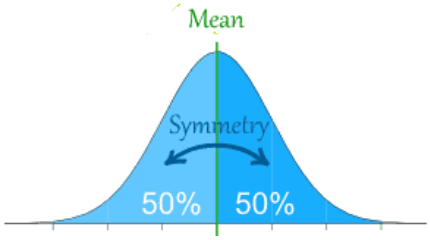
where

$e \approx 2.718$ ($\approx$ means "is approximately equal to")

$\pi \approx 3.14$

μ = population mean

σ = population standard deviation



PROPERTIES OF NORMAL DISTRIBUTION

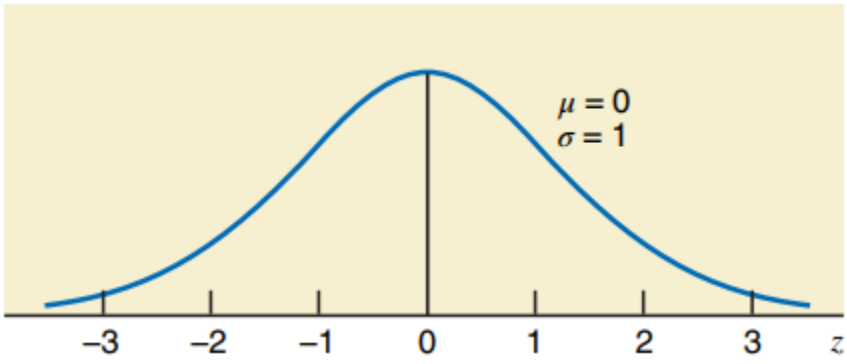
1. Mean of Normal Distribution is $E(X) = \mu$
2. The Variance of Normal Distribution is $V(X) = \sigma^2$
3. It is a bell –shaped distribution.
4. It is symmetric about the mean.
5. It extends from $-\infty$ *to* $+\infty$
6. The total area under the curve is 1.
7. A normal distribution curve is unimodal.
8. In Normal Distribution $Mean = Median = Mode$
9. The curve is continuous; that is, there are no gaps or holes.
10. The curve never touches the *x*-axis.It approaches the x-axis but never crosses it.

STANDARD NORMAL DISTRIBUTION

If Normal distribution have mean 0 and standard deviation 1 than it is denoted by $Z \sim N(0,1)$. In this case the probability density function is given by

$$f(z)=\frac{1}{\sqrt{2\pi}}e^{-\frac{z^2}{2}} \quad -\infty < z < \infty$$

The standard normal distribution curve.



FINDING PROBABILITIES USING STANDARD NORMAL DISTRIBUTION

If we have a normal distribution having mean = 0 and variance = 1.
ie $Z \sim N(0,1)$. Then we can find the area using the formulas given below.

- $P(Z<a)=\Phi(a)$
- $P(Z>a)=1-P(Z<a)=1-\Phi(a)$
- $P(Z<-a)=\Phi(-a)$ *or* $P(Z<-a)=P(Z>a)=1-\Phi(a)$
- $P(Z>-a)=1-P(Z<-a)=1-\Phi(-a)$ *or* $P(Z>-a)=P(Z<a)=\Phi(a)$
- $P(a<Z<b)=\Phi(b)-\Phi(a)$
- $P(-a<Z<b)=\Phi(b)+\Phi(-a)=\Phi(a)+\Phi(b)-1$
- $P(-a<Z<-b)=\Phi(-b)-\Phi(-a)=\Phi(a)-\Phi(b)$
- $P(|z|<a)=P(-a<Z<a)=\Phi(a)-\Phi(-a)=2\Phi(a)-1$
- $P(|z|>a)=1-P(|z|<a)=2(1-\Phi(a))$

EXAMPLES OF STANDARD NORMAL DISTRIBUTION

EXAMPLE-1 If $Z \sim N(0,1)$ then find the following probabilities

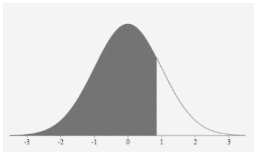
$(a)\ P(Z < 0.85)$ $(b)\ P(Z > 1.72)$

SOLUTION *Here* $Z \sim N(0,1)$

$(a)\ P(Z < 0.85)$

$= \Phi(0.85)$

$= 0.8023$



$(b)\ P(Z > 1.72)$

$= 1 - P(Z < 1.72)$

$= 1 - \Phi(1.72)$

$= 1 - 0.9573 = 0.0427$



EXAMPLE-2 If $Z \sim N(0,1)$ then find the following probabilities

$(a)\ P(Z < -1.37)$ $(b)\ P(Z > -2.14)$

SOLUTION *Here* $Z \sim N(0,1)$

$(a)\ P(Z < -1.37)$

$= \Phi(-1.37)$

$= 0.0853$

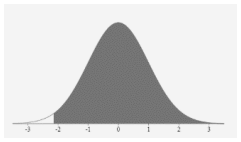


$(b)\ P(Z > -2.14)$

$= 1 - P(Z < -2.14)$

$= 1 - \Phi(-2.14)$

$= 1 - 0.0162 = 0.9838$



EXAMPLE-3 If $Z \sim N(0,1)$ then find the following probabilities

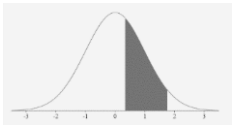
$(a)\ P(0.34 < Z < 1.75)$ $(b)\ P(-1.4 < Z < -0.6)$ $(c)\ P(-2.69 < Z < 1.86)$

SOLUTION *Here* $Z \sim N(0,1)$

$(a)\ P(0.34 < Z < 1.75)$

$= \Phi(1.75) - \Phi(0.34)$

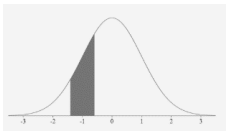
$= 0.9599 - 0.6331 = 0.3268$



$(b)\ P(-1.4 < Z < -0.6)$

$= \Phi(-0.6) - \Phi(-1.4)$

$= 0.2743 - 0.0808 = 0.1935$



$(c)\ P(-2.69 < Z < 1.86)$

$= \Phi(1.86) - \Phi(-2.69)$

$= 0.9686 - 0.0036 = 0.96$



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EXAMPLE-4 If $Z \sim N(0,1)$ then find the value of ***a*** if

$(a)\ P(Z < a) = 0.9693$

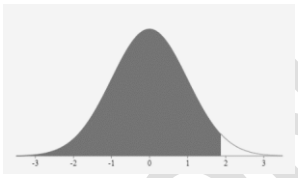
$(b)\ P(Z > a) = 0.3802$

$(c)\ P(Z < a) = 0.0793$

$(d)\ P(Z > a) = 0.7367$

SOLUTION *Here* $Z \sim N(0,1)$

$(a)\ P(Z < a) = 0.9693$

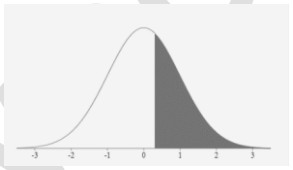


$\Phi(a) = 0.9693$

$a = \Phi^{-1}(0.9693)$

$a = 1.87$

$(b)\ P(Z > a) = 0.3802$



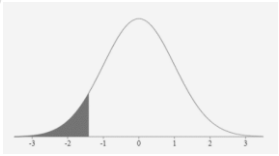
$1 - P(Z < a)) = 0.3802$

$1 - \Phi(a) = 0.3802$

$\Phi(a) = 1 - 0.3802 = 0.6198$

$a = \Phi^{-1}(0.6198) = 0.305$

$(c)\ P(Z < a) = 0.0793$

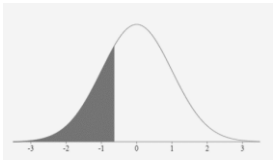


$\Phi(a) = 0.0793$

$a = \Phi^{-1}(0.9693)$

$a = -1.41$

$(d)\ P(Z > a) = 0.7367$



$1 - P(Z < a)) = 0.7367$

$1 - \Phi(a) = 0.7367$

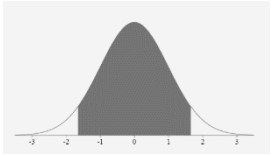
$\Phi(a) = 1 - 0.7367 = 0.2633$

$a = \Phi^{-1}(0.2633) = -0.633$

EXAMPLE-5 If $Z \sim N(0,1)$ then find the value of ***a*** if *if* $P(|z| < a) = 0.9$

SOLUTION *Here* $Z \sim N(0,1)$

$P(-a < Z < a) = 0.9$



$\Phi(a) - \Phi(-a) = 0.9$

$\Phi(a) - [1 - \Phi(a)] = 0.9$

$\Phi(a) - 1 + \Phi(a) = 0.9$

$2\Phi(a) = 1.9$

$\Phi(a) = 0.95$

$a = \Phi^{-1}(0.95) = 1.645$

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GENERAL NORMAL DISTRIBUTION

If Normal distribution have mean μ (any value) and standard deviation σ (any value) than it is denoted by $X \sim N(\mu, \sigma^2)$. In this case the probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2\sigma^2}(x-\mu)^2} \quad -\infty < x < \infty$$

FINDING PROBABILITIES USING GENERAL NORMAL DISTRIBUTION

If the given normal distribution is not standard. Then first we convert it to standard normal distribution and then solve the problem.
Convert $X \sim N(\mu, \sigma^2) \rightarrow Z \sim N(0,1)$ using the formula

$$Z = \frac{X - \mu}{\sigma}$$

Where $X = Normal Variable$

$\mu = Mean$

$\sigma^2 = Variance$

Probability Calculations for Normal Distributions

If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

The random variable Z is known as the "standardized" version of the random variable X. This result implies that the probability values of a general normal distribution can be related to the cumulative distribution function of the standard normal distribution $\Phi(x)$ through the relationship

$$P(a \le X \le b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

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EXAMPLES OF GENERAL NORMAL DISTRIBUTION

EXAMPLE-1 If $X \sim N(-8,12)$ then find the following probabilities

$(a)\ P(X < -9.8)$

$(b)\ P(X > -8.2)$

$(c)\ P(-7 < X < 0.5)$

SOLUTION *Here* $X \sim N(-8,12) \Rightarrow \mu = -8,\ \sigma = \sqrt{12}$

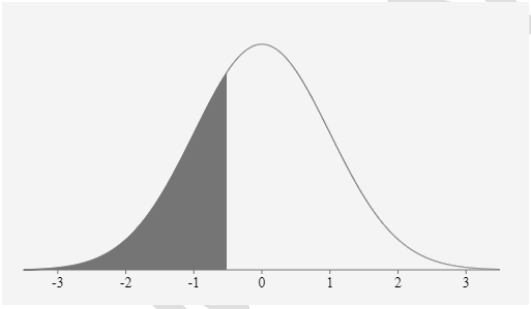
$(a)\ P(X < -9.8) = ?$

$$Z = \frac{X-\mu}{\sigma} = \frac{-9.8+8}{\sqrt{12}} = -0.52$$

$So\ P(Z < -0.52)$

$= \Phi(-0.52)$

$= 0.3015$



$(b)\ P(X > -8.2) = ?$

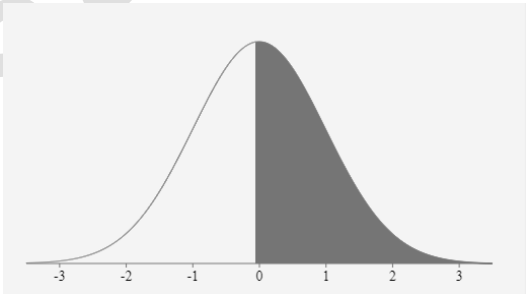
$$Z = \frac{X-\mu}{\sigma} = \frac{-8.2+8}{\sqrt{12}} = -0.06$$

$So\ P(Z > -0.06)$

$= 1 - P(Z < -0.06)$

$= 1 - \Phi(-0.06) = 1 - 0.4761$

$= 0.5239$



$(c)\ P(-7 < X < 0.5) = ?$

$$Z_1 = \frac{x-\mu}{\sigma} = \frac{-7+8}{\sqrt{12}} = 0.29$$

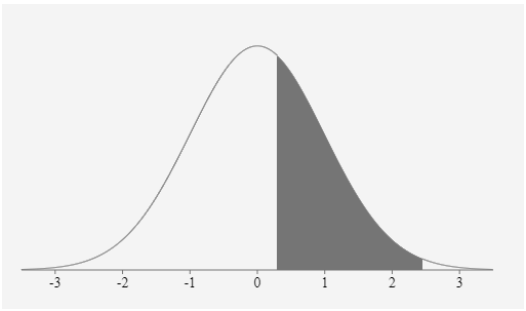
$$Z_2 = \frac{x-\mu}{\sigma} = \frac{0.5+8}{\sqrt{12}} = 2.45$$

$So\ P(0.29 < Z < 2.45)$

$= \Phi(2.45) - \Phi(0.29)$

$= 0.9929 - 0.6141$

$= 0.3788$



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EXAMPLE-2 If $X \sim N(100,81)$ then find the following probabilities

$(a)\ P(|X-100|<18)$

$(b)\ P(|X-100|>5)$

$(c)\ P(12<X-100<15)$

SOLUTION *Here* $X \sim N(100,81) \Rightarrow \mu = 100,\ \sigma = 9$

$(a)\ P(|X-100|<18)$

$= P\left(\left|\frac{X-100}{9}\right| < \frac{18}{9}\right)$

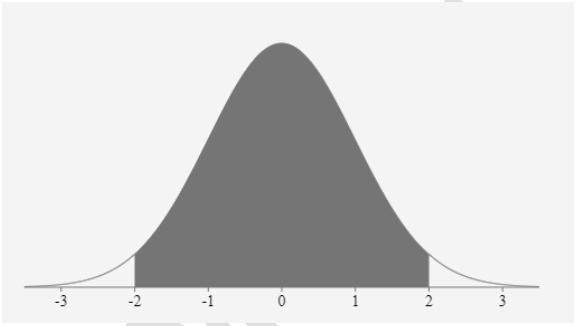
$= P(|Z|<2)$

$= P(-2<Z<2)$

$= \Phi(2) - \Phi(-2)$

$= 0.9772 - 0.0228$

$= 0.9544$



$(b)\ P(|X-100|>5)$

$= P\left(\left|\frac{X-100}{9}\right| > \frac{5}{9}\right)$

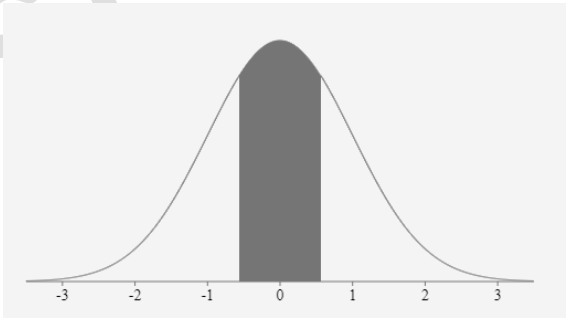
$= P(|Z|>0.56) = 1 - P(|Z|<0.56)$

$= 1 - P(-0.56<Z<0.56)$

$= 1 - [\Phi(0.56) - \Phi(-0.56)]$

$= 1 - [0.7123 - 0.2877]$

$= 0.5754$



$(c)\ P(12<X-100<15)$

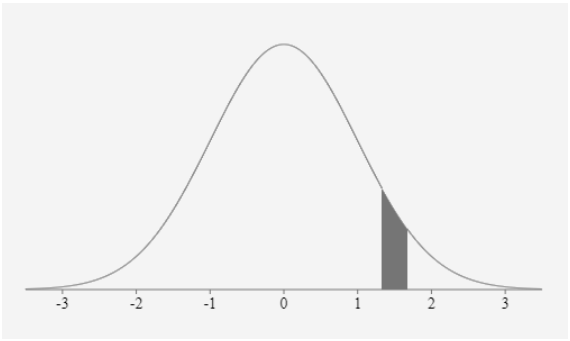
$= P\left(\frac{12}{9} < \frac{X-100}{9} < \frac{15}{9}\right)$

$= P(1.33<Z<1.67)$

$= \Phi(1.67) - \Phi(1.33)]$

$= 0.9525 - 0.9082$

$= 0.0443$



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EXAMPLE-3 If $X \sim N(200,36)$ and *if* $P(X > a) = 0.9386$ then find the value of ***a***.

SOLUTION *Here* $X \sim N(200,36) \Rightarrow \mu = 200, \quad \sigma = 6$

Now $P(X > a) = 0.9386$

$$Z = \frac{a-200}{6} = b \ (say)$$

So $P(Z > b) = 0.9386$

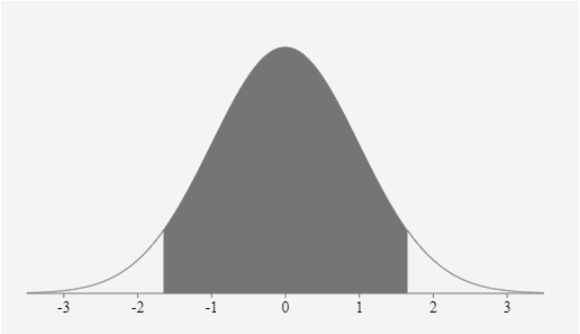
$1 - P(Z < b)) = 0.9386$

$1 - \Phi(b) = 0.9386$

$\Phi(b) = 1 - 0.9386 = 0.0614$

$b = \Phi^{-1}(0.0614) = -1.50$

$$So \ \frac{a-200}{6} = -1.50 \Rightarrow a = 191$$



EXAMPLE-4 If $X \sim N(10,2)$ and *if* $P(|X - 10| \le a) = 0.9662$ then find the value of ***a***.

SOLUTION *Here* $X \sim N(10,2) \Rightarrow \mu = 10, \ \sigma = \sqrt{2}$

Now $P(|X - 10| \le a) = 0.9662$

$$P\left(\left|\frac{X-10}{\sqrt{2}}\right| \le \frac{a}{\sqrt{2}}\right) = 0.9662$$

$P(|Z| \le a/\sqrt{2}) = 09662$

$P(|Z| \le b) = 0.9662 \quad where \quad b = a/\sqrt{2}$

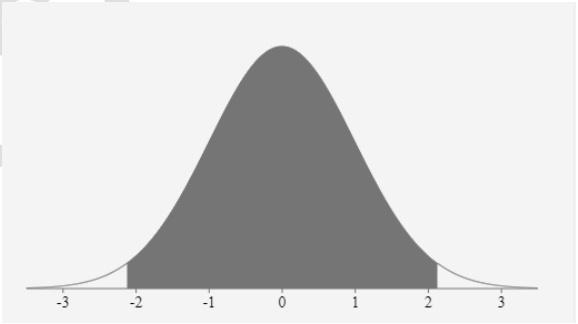
$P(-b \le Z \le b) = 0.9662$

$\Phi(b) - \Phi(-b) = 0.9662$

$2\Phi(b) - 1 = 0.9662 \Rightarrow \Phi(b) = 0.9831$

$b = \Phi^{-1}(0.9831) = 2.12$

$So \ \ a/\sqrt{2} = 2.12 \quad \Rightarrow \quad a = 3$



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EXAMPLE-5 If $X \sim N(4.3, 0.12^2)$ then find the following values

(a) *Lower and Upper Quartiles*

(b) $95th$ *Percentile*

(c) *value of* c *for which* $P(4.3-c<X<4.3+c)=0.8$

SOLUTION *Here* $X \sim N(4.3, 0.12^2)$, *So* $\mu = 4.3$ *and* $\sigma = 0.12$

(a) *For Lower Quartile* $= Q_1$,

Solve $P(X \le Q_1) = 0.25$ *for* Q_1

$$Z = \frac{Q_1 - 4.3}{0.12} = a \ (say)$$

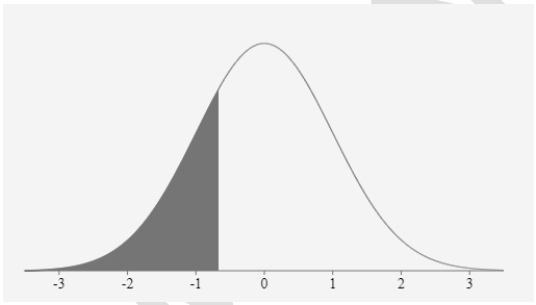
$$P(Z \le a) = 0.25$$

$$\Phi(a) = 0.25$$

$$a = \Phi^{-1}(0.25) = -0.67$$

$$So \ \frac{Q_1 - 4.3}{0.12} = -0.67$$

$$\Rightarrow Q_1 = 4.2196$$



(a) *For Upper Quartile* $= Q_3$,

Solve $P(X \le Q_3) = 0.75$ *for* Q_3

$$Z = \frac{Q_3 - 4.3}{0.12} = b \ (say)$$

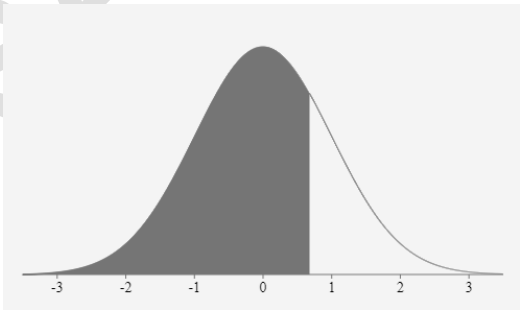
$$P(Z \le b) = 0.75$$

$$\Phi(b) = 0.75$$

$$b = \Phi^{-1}(0.75) = 0.67$$

$$So \ \frac{Q_3 - 4.3}{0.12} = 0.67$$

$$\Rightarrow Q_3 = 4.3804$$



(b) *For* $95th$ *Percentile* $= P_{95}$,

Solve $P(X \le P_{95}) = 0.95$ *for* P_{95}

$$Z = \frac{P_{95} - 4.3}{0.12} = c \ (say)$$

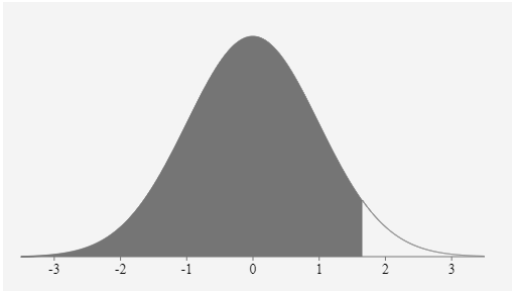
$$P(Z \le c) = 0.95$$

$$\Phi(c) = 0.95$$

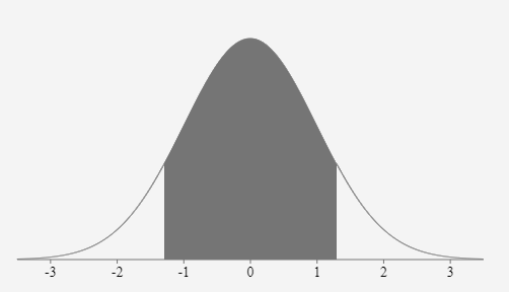
$$c = \Phi^{-1}(0.95) = 1.65$$

$$So \ \frac{P_{95} - 4.3}{0.12} = 1.65$$

$$\Rightarrow P_{95} = 4.498$$



(*c*) *Here* $P(4.3-c < X < 4.3+c) = 0.8$



*Converting to s*tan*dard version*

$u\sin g\ Z = \dfrac{X-\mu}{\sigma}$, *we get*

$$P\left(\frac{(4.3-c)-4.3}{0.12} < Z < \frac{(4.3+c)-4.3}{0.12}\right) = 0.8$$

$$P\left(\frac{4.3-c-4.3}{0.12} < Z < \frac{4.3+c-4.3}{0.12}\right) = 0.8$$

$$P\left(\frac{-c}{0.12} < Z < \frac{c}{0.12}\right) = 0.8$$

$$\Phi\left(\frac{c}{0.12}\right) - \Phi\left(\frac{-c}{0.12}\right) = 0.8$$

$$\Phi\left(\frac{c}{0.12}\right) - [1 - \Phi\left(\frac{c}{0.12}\right)] = 0.8$$

$$\Phi\left(\frac{c}{0.12}\right) - 1 + \Phi\left(\frac{c}{0.12}\right) = 0.8$$

$$2\Phi\left(\frac{c}{0.12}\right) = 1.8$$

$$\Phi\left(\frac{c}{0.12}\right) = 0.9$$

$$\frac{c}{0.12} = \Phi^{-1}(0.9)$$

$$\frac{c}{0.12} = 1.29$$

$$c = 0.1548$$

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APPLICATIONS OF NORMAL DISTRIBUTION

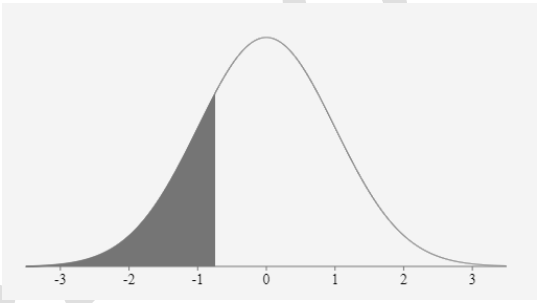
EXAMPLE-1 The life of a certain make of electric bulbs is normally distributed with a mean life of 2000 hours and a standard deviation of 120 hours. Estimate the probability that the life of such bulb will be
(a) lesser then 1910 hours
(b) greater then 2150 hours
(c) within the range of 1850 hours to 2090 hours.

SOLUTION *Let X denote life of such bulbs.*
Here $X \sim N(2000,14400) \Rightarrow \mu = 2000,\ \sigma = 120$

$(a)\ P(X < 1910) = ?$

$$Z = \frac{X-\mu}{\sigma} = \frac{1910-2000}{120} = -0.75$$

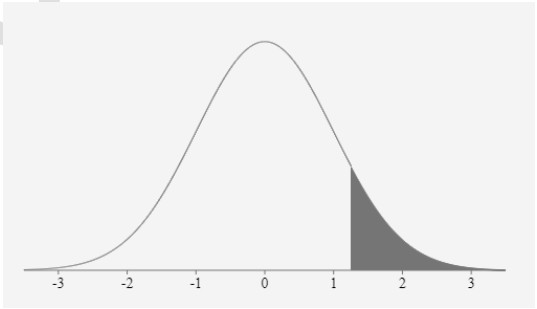
$$So\ P(Z < -0.75)$$
$$= \Phi(-0.75)$$
$$= 0.2266$$



$(b)\ P(X > 2150) = ?$

$$Z = \frac{X-\mu}{\sigma} = \frac{2150-2000}{120} = 1.25$$

$$So\ P(Z > 1.25)$$
$$= 1 - P(Z < 1.25)$$
$$= 1 - \Phi(1.25) = 1 - 0.8944$$
$$= 0.1056$$

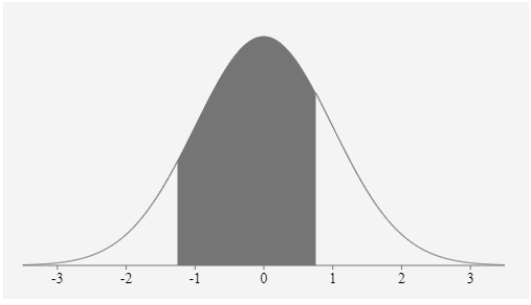


$(c)\ P(1850 < X < 2090) = ?$

$$Z_1 = \frac{X-\mu}{\sigma} = \frac{1850-2000}{120} = -1.25$$

$$Z_2 = \frac{X-\mu}{\sigma} = \frac{2090-2000}{120} = 0.75$$

$$So\ P(-1.25 < Z < 0.75)$$
$$= \Phi(0.75) - \Phi(-1.25)$$
$$= 0.7734 - 0.1056$$
$$= 0.6678$$



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EXAMPLE-2 Lengths of metal strips produced by a machine are normally distributed with mean length of 150 cm and a standard deviation of 10 cm. Find the probability that the length of a randomly selected strip is
(a) shorter than 165 cm.
(b) longer than 160 cm
(c) within 5 cm of the mean.

SOLUTION *Let X denote length of metal strips.*

Here $X \sim N(150,100) \Rightarrow \mu = 150,\ \sigma = 10$

$(a)\ P(X < 165) = ?$

$$Z = \frac{X-\mu}{\sigma} = \frac{165-150}{10} = 1.5$$

$So\ P(Z < 1.5)$

$= \Phi(1.5)$

$= 0.93$



$(b)\ P(X > 160) = ?$

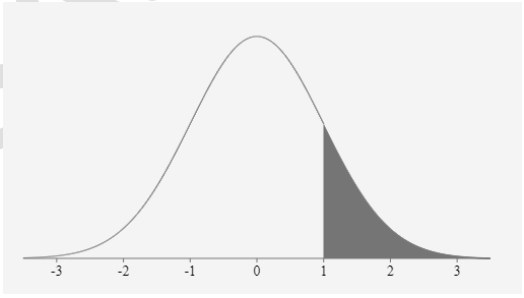
$$Z = \frac{X-\mu}{\sigma} = \frac{160-150}{10} = 1$$

$So\ P(Z > 1)$

$= 1 - P(Z < 1)$

$= 1 - \Phi(1)$

$= 1 - 0.8413 = 0.1587$



$(c)\ P(145 < X < 155) = ?$

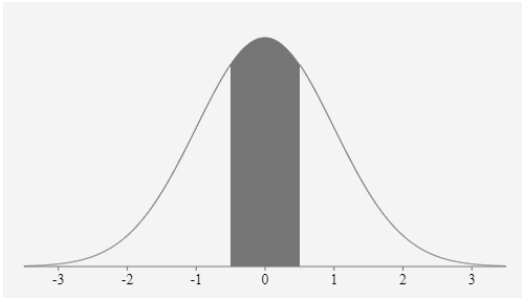
$$Z_1 = \frac{X-\mu}{\sigma} = \frac{145-150}{10} = -0.5$$

$$Z_2 = \frac{X-\mu}{\sigma} = \frac{155-150}{10} = 0.5$$

$So\ P(-0.5 < Z < 0.5)$

$= \Phi(0.5) - \Phi(-0.5)$

$= 0.6915 - 0.3085 = 0.383$



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EXAMPLE-3 Time taken by the milk man to deliver milk to the high street is normally distributed with mean of 12 minutes and a standard deviation of 2 minutes. He delivers milk every day. Estimate number of days during the year when he takes
(a) longer than 17 minutes
(b) less than 10 minutes
(c) between 9 and 13 minutes.

SOLUTION *Let* X *denote time in* min *utes to deliver milk to High Street.*

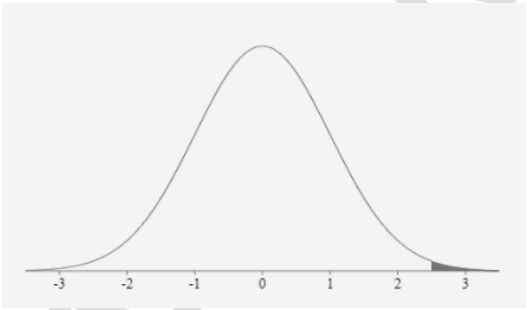
Here $X \sim N(12,4) \Rightarrow \mu = 12,\ \sigma = 2$

$(a)\ P(X > 17) = ?$

$$Z = \frac{X-\mu}{\sigma} = \frac{17-12}{2} = 2.5$$

$$So\ \ P(Z > 2.5)$$
$$= 1 - P(Z < 2.5)$$
$$= 1 - \Phi(2.5) = 1 - 0.9938$$
$$= 0.0062$$

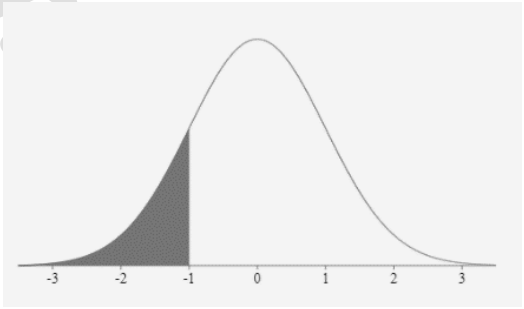
So number of such days in a year $= 365 \times 0.0062 = 2.263 \equiv 2\,days$



$(b)\ P(X < 10) = ?$

$$Z = \frac{X-\mu}{\sigma} = \frac{10-12}{2} = -1$$

$$So\ \ P(Z < -1)$$
$$= \Phi(-1) = 0.1587$$



So number of such days in a year $= 365 \times 0.1587 = 57.92 \equiv 58\,days$

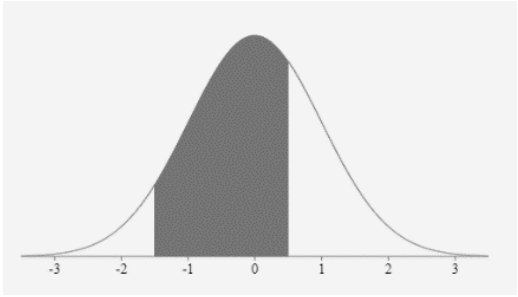
$(c)\ P(9 < X < 13) = ?$

$$Z_1 = \frac{X-\mu}{\sigma} = \frac{9-12}{2} = -1.5$$

$$Z_2 = \frac{X-\mu}{\sigma} = \frac{13-12}{2} = 0.5$$

$$So\ \ P(-1.5 < Z < 0.5)$$
$$= \Phi(0.5) - \Phi(-1.5)$$
$$= 0.6915 - 0.0668 = 0.6247$$

So number of such days in a year $= 365 \times 0.6247 = 228.01 \equiv 228\,days$

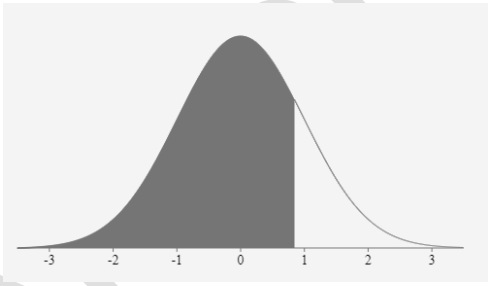


EXAMPLE-4 The heights of female students at particular college are normally distributed with a mean of 169 cm and a standard deviation of 9 cm.
(a) Given that 80% of these female students have a height less than $h\ cm$, find the value of h.
(b) Given that 60% of these female students have a height greater than $g\ cm$, find the value of g.

SOLUTION *Let X denote height of female students in cm.*

$$Here\ X \sim N(169,81) \Rightarrow \mu = 169,\ \sigma = 9$$

$$(a)\ P(X < h) = 0.8$$



$$Z = \frac{h-169}{9} = a\ (say)$$

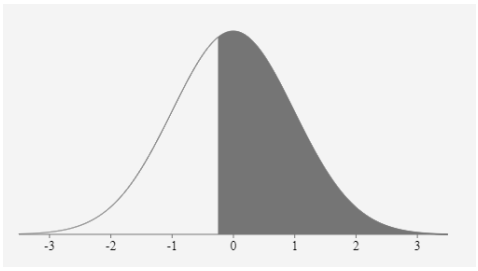
$$So\ P(Z < a) = 0.8$$

$$\Phi(a) = 0.8$$

$$a = \Phi^{-1}(0.8) = 0.842$$

$$So\ \frac{h-169}{9} = 0.84 \Rightarrow h = 176.58\,cm$$

$$(b)\ P(X > g) = 0.6$$



$$Z = \frac{g-169}{9} = b\ (say)$$

$$So\ P(Z > b) = 0.6$$

$$1 - P(Z < b) = 0.6$$

$$1 - \Phi(b) = 0.6$$

$$\Phi(b) = 0.4$$

$$b = \Phi^{-1}(0.4) = -0.25$$

$$So\ \frac{g-169}{9} = -0.25 \Rightarrow g = 166.72\,cm$$

EXAMPLE-5 The marks of 500 candidates in an examination are normally distributed with a mean of 45 marks and a standard deviation of 20 marks.
(a) Given that the pass marks is 41, estimate the number of candidates who passed the examination.
(b) If 5% candidates obtain distinction by scoring ***a*** marks or more, estimate the value of a.
(c) estimate interquartile range of the distribution.

SOLUTION *Let X denote marks of the candidates in the exa*min*ation.*

$$Here\ X \sim N(45,400) \Rightarrow \mu = 45,\ \sigma = 20$$

$(a)\ P(X \geq 41) = ?$

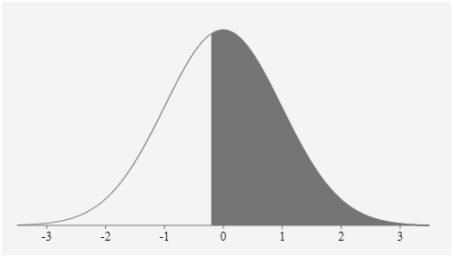
$$Z = \frac{X-\mu}{\sigma} = \frac{41-45}{20} = -0.2$$

$So\ P(Z \geq -0.2)$

$= 1 - P(Z < -0.2)$

$= 1 - \Phi(-0.2) = 1 - 0.4207$

$= 0.5793$



So number of candidates who pass $= 500 \times 0.5793 = 289.65 \equiv 290$

$(b)\ P(X \geq a) = 0.05$

$$Z = \frac{a-45}{20} = b\ (say)$$

$So\ P(Z \geq b) = 0.05$

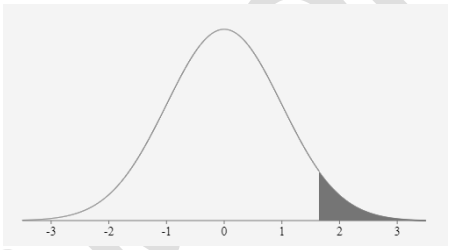
$1 - P(Z < b) = 0.05$

$1 - \Phi(b) = 0.05$

$\Phi(b) = 0.95$

$b = \Phi^{-1}(0.95) = 1.645$

$$So\ \frac{a-45}{20} = 1.645 \Rightarrow a = 78$$



$(a)\ Inter\ Quartile\ Range = Q_3 - Q_1$

$Solve\ P(X \leq Q_1) = 0.25\ \ for\ Q_1$

$$Z = \frac{Q_1-45}{20} = a\ (say)$$

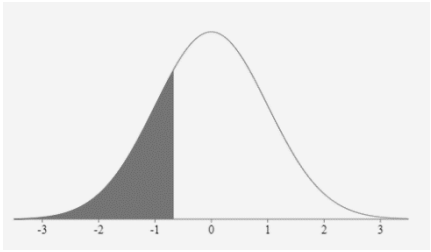
$P(Z \leq a) = 0.25$

$\Phi(a) = 0.25$

$a = \Phi^{-1}(0.25) = -0.67$

$$So\ \frac{Q_1-45}{20} = -0.67$$

$\Rightarrow\ Q_1 = 31.6$



$Solve\ P(X \leq Q_3) = 0.75\ \ for\ Q_3$

$$Z = \frac{Q_3-45}{20} = b\ (say)$$

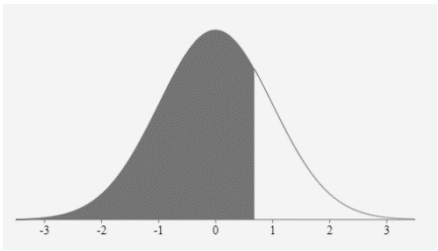
$P(Z \leq b) = 0.75$

$\Phi(b) = 0.75$

$b = \Phi^{-1}(0.75) = 0.67$

$$So\ \frac{Q_3-45}{20} = 0.67$$

$\Rightarrow\ Q_3 = 58.4$



So Interquartile range $= Q_3 - Q_1 = 58.4 - 31.6 = 26.8$

IMPORTANT CONCEPTS AND FUNCTIONS IN PROBABILITY

DEFINITE INTEGRAL

$$\int_a^b f(x)dx = F(b) - F(a)$$

PROPERTIES OF DEFINITE INTEGRAL

1. $\int_a^a f(x)dx = 0$
2. $\int_a^b f(x)dx = -\int_b^a f(x)dx$
3. $\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$
4. $$\int_{-a}^{a} f(x)dx = \begin{cases} 0, & \text{if } f(x) \text{ is an odd function} \\ 2\int_0^a f(x)dx, & \text{if } f(x) \text{ is an even function} \end{cases}$$

GAMMA FUNCTION

$$\Gamma(n) = \int_0^\infty x^{n-1}e^{-x}dx \quad \text{also} \quad \int_0^\infty x^{n-1}e^{-ax}dx = \frac{\Gamma(n)}{a^n}$$

PROPERTIES OF GAMMA FUNCTION

1. $\Gamma(n) = (n-1).\Gamma(n-1)$
2. $\Gamma(n) = (n-1)!$ *If 'n' is a positive integer.*
3. $\Gamma(1) = 1$
4. $\Gamma(\frac{1}{2}) = \sqrt{\pi}$
5. $\Gamma(-\frac{1}{2}) = \frac{1}{\sqrt{\pi}}$

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PROPERTIES OF NORMAL DISTRIBUTION

PROPERT-1 AREA UNDER THE NORMAL CURVE

We have to show that $\int_{-\infty}^{\infty} f(x)dx = 1$

$$L.H.S = \int_{-\infty}^{\infty} f(x)dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx ------(1)$$

$$Let \ \frac{x-\mu}{\sigma} = y \ \Rightarrow x = \mu + \sigma y \ \Rightarrow dx = \sigma\, dy$$

when $x = -\infty, y = -\infty$, *when* $x = \infty, y = \infty$, *So equation* (1) *becomes*

$$L.H.S = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} \sigma\, dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} dy$$

$$= \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-\frac{1}{2}y^2} dy ----(2) \quad (As \ \text{int}egrand \ is \ an \ even \ function)$$

$$Let \ \ y^2 = z \ \Rightarrow 2y\, dy = dz \ \Rightarrow dy = \frac{1}{2y} dz \ \Rightarrow dy = \frac{1}{2\sqrt{z}} dz$$

when $y = 0, z = 0$, *when* $y = \infty, z = \infty$, *So equation* (2) *becomes*

$$L.H.S = \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-\frac{1}{2}z} \frac{1}{2\sqrt{z}} dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} z^{-\frac{1}{2}} e^{-\frac{1}{2}z} dz$$

$$= \frac{1}{\sqrt{2\pi}} \times \frac{\Gamma(1/2)}{(1/2)^{1/2}} = \frac{1}{\sqrt{2\pi}} \times \frac{\sqrt{2} \times \sqrt{\pi}}{1} = 1 = R.H.S$$

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PROPERT-2 MEAN OF NORMAL DISTRIBUTION

$$Mean = E(X) = \int_{-\infty}^{\infty} x\, f(x)dx$$

$$= \int_{-\infty}^{\infty} x \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx ------(1)$$

Let $\frac{x-\mu}{\sigma} = y \Rightarrow x = \mu + \sigma y \Rightarrow dx = \sigma\, dy$

when $x = -\infty, y = -\infty$, *when* $x = \infty, y = \infty$, *So equation* (1) *becomes*

$$E(X) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma y) e^{-\frac{1}{2}y^2} \sigma\, dy$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} dy + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-\frac{1}{2}y^2} dy$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} dy + \frac{\sigma}{\sqrt{2\pi}}(0) \ (As \text{ int } egrand \text{ is an odd function})$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} dy$$

$$= \mu \times \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy = \mu \times (1) = \mu \qquad \left(As \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} = 1\right)$$

PROPERT-3 VARIANCE OF NORMAL DISTRIBUTION

$$V(X) = E(X^2) - (E(X))^2 --------(1)$$

we have $E(X) = \mu$

Now $E(X^2) = \int_{-\infty}^{\infty} x^2\, f(x)dx$

$$= \int_{-\infty}^{\infty} x^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx ----(2)$$

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Let $\frac{x-\mu}{\sigma} = y \Rightarrow x = \mu + \sigma y \Rightarrow dx = \sigma\, dy$

when $x = -\infty, y = -\infty$, *when* $x = \infty, y = \infty$, *So equation* (2) *becomes*

$$E\left(X^2\right) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\mu + \sigma y\right)^2 e^{-\frac{1}{2}y^2}\, \sigma\, dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\mu^2 + 2\sigma y + \sigma^2 y^2\right) e^{-\frac{1}{2}y^2}\, \sigma\, dy$$

$$= \frac{\mu^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2}\, dy + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(2\sigma y + \sigma^2 y^2\right) e^{-\frac{1}{2}y^2}\, dy$$

$$= \mu^2 \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2}\, dy + \frac{2\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y\, e^{-\frac{1}{2}y^2}\, dy + \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-\frac{1}{2}y^2}\, dy$$

$$= \mu^2(1) + \frac{2\sigma}{\sqrt{2\pi}}(0) + \frac{2\sigma^2}{\sqrt{2\pi}} \int_0^{\infty} y^2 e^{-\frac{1}{2}y^2}\, dy$$

$$= \mu^2 + \frac{2\sigma^2}{\sqrt{2\pi}} \int_0^{\infty} y^2 e^{-\frac{1}{2}y^2}\, dy \text{ ---------(3)}$$

Let $y^2 = z \Rightarrow 2y\, dy = dz \Rightarrow dy = \frac{1}{2y} dz \Rightarrow dy = \frac{1}{2\sqrt{z}} dz$

when $y = 0, z = 0$, *when* $y = \infty, z = \infty$, *So equation* (3) *becomes*

$$L.H.S = \mu^2 + \frac{2\sigma^2}{\sqrt{2\pi}} \int_0^{\infty} z\, e^{-\frac{1}{2}z} \frac{1}{2\sqrt{z}} dz$$

$$= \mu^2 + \frac{\sigma^2}{\sqrt{2\pi}} \int_0^{\infty} z^{\frac{1}{2}} e^{-\frac{1}{2}z}\, dz$$

$$= \mu^2 + \frac{\sigma^2}{\sqrt{2\pi}} \times \frac{\Gamma(3/2)}{(1/2)^{3/2}}$$

$$= \mu^2 + \frac{\sigma^2}{\sqrt{2\pi}} \times \frac{(1/2)\Gamma(1/2)}{(1/2)^1 (1/2)^{1/2}} = \mu^2 + \frac{\sigma^2}{\sqrt{2\pi}} \times \frac{(1/2)\times\sqrt{2\pi}}{(1/2)} = \mu^2 + \sigma^2$$

Put the values of $E(X)$ *and* $E\left(X^2\right)$ *in equation* (1), *we get*

$$V(X) = \mu^2 + \sigma^2 - (\mu)^2 = \sigma^2$$

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PROPERT-4 MOMENT GENERATING FUNCTION

Moment Generating Function is given by

$$M_X(t) = E\left(e^{tx}\right) = \int_{-\infty}^{\infty} e^{tx}\, f(x)\,dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx}\; e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\, dx \text{ ------(1)}$$

Let $\frac{x-\mu}{\sigma} = y \Rightarrow x = \mu + \sigma y \Rightarrow dx = \sigma\, dy$

when $x = -\infty, y = -\infty$, *when* $x = \infty, y = \infty$, *So equation* (1) *becomes*

$$M_X(t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\mu+\sigma y)}\; e^{-\frac{1}{2}y^2}\, \sigma\, dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\mu t + t\sigma y}\; e^{-\frac{1}{2}y^2}\, dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\mu t} e^{t\sigma y}\; e^{-\frac{1}{2}y^2}\, dy$$

$$= \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} e^{t\sigma y}\, dy = \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2 + t\sigma y}\, dy = \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y^2 - 2t\sigma y)}\, dy$$

$$= \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y^2 - 2t\sigma y + t^2\sigma^2 - t^2\sigma^2)}\, dy$$

$$= \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y^2 - 2t\sigma y + t^2\sigma^2)}\, e^{-\frac{1}{2}(-t^2\sigma^2)}\, dy$$

$$= \frac{e^{\mu t} \times e^{-\frac{1}{2}(-t^2\sigma^2)}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y^2 - 2t\sigma y + t^2\sigma^2)}\, dy$$

$$= \frac{e^{\mu t} \times e^{\frac{1}{2}(t^2\sigma^2)}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y - t\sigma)^2}\, dy$$

$$M_X(t) = \frac{e^{\mu t + \frac{1}{2}t^2\sigma^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y - t\sigma)^2}\, dy \text{ -------- (2)}$$

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Let $y-t\sigma=z \Rightarrow dy=dz$

when $y=-\infty, z=-\infty,$ *when* $x=\infty, z=\infty$

So equation (2) *becomes*

$$M_X(t)=\frac{e^{\mu t+\frac{1}{2}t^2\sigma^2}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}z^2}\,dz=e^{\mu t+\frac{1}{2}t^2\sigma^2}\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}\,dz=e^{\mu t+\frac{1}{2}t^2\sigma^2} \quad (1)$$

$$M_X(t)=e^{\mu t+\frac{1}{2}t^2\sigma^2}$$

Mean using Moment Generating Function is given by

$$E(X)=\left[\frac{d}{dt}M_X(t)\right]_{t=0}$$

$$=\left[\frac{d}{dt}\left(e^{\mu t+\frac{1}{2}t^2\sigma^2}\right)\right]_{t=0}$$

$$=\left[(\mu(1)+\frac{1}{2}(2t)\sigma^2)\times e^{\mu t+\frac{1}{2}t^2\sigma^2}\right]_{t=0}$$

$$=\left[(\mu+t\sigma^2)e^{\mu t+\frac{1}{2}t^2\sigma^2}\right]_{t=0}=\mu$$

Variance using Moment Generating Function is given by

For variance we shall first find $E(X^2)$

$$E(X^2)=\left[\frac{d^2}{dt^2}M_X(t)\right]_{t=0}$$

$$=\left[\frac{d}{dt}\left((\mu+t\sigma^2)e^{\mu t+\frac{1}{2}t^2\sigma^2}\right)\right]_{t=0}$$

$$=\left[\sigma^2\times e^{\mu t+\frac{1}{2}t^2\sigma^2}+(\mu+t\sigma^2)e^{\mu t+\frac{1}{2}t^2\sigma^2}\times(\mu+t\sigma^2)\right]_{t=0}$$

$$=\sigma^2+\mu^2$$

Now $V(X)=E(X^2)-(E(X))^2$

$$=\sigma^2+\mu^2-\mu^2=\sigma^2$$

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PROPERT-5 CUMULANT GENERATING FUNCTION

Cumulant Generating Function is given by

$$K_X(t) = \ln(M_X(t))$$

$$= \ln\left(e^{\mu t + \frac{1}{2}t^2\sigma^2}\right)$$

$$= (\mu t + \frac{1}{2}t^2\sigma^2)\ln e$$

$$= (\mu t + \frac{1}{2}t^2\sigma^2)(1)$$

$$= \mu t + \frac{1}{2}t^2\sigma^2$$

Mean using Cumulant Generating Function is given by

$$E(X) = \left[\frac{d}{dt}K_X(t)\right]_{t=0}$$

$$= \left[\frac{d}{dt}\left(\mu t + \frac{1}{2}t^2\sigma^2\right)\right]_{t=0}$$

$$= \left[\mu + \frac{1}{2}\times 2t\sigma^2\right]_{t=0}$$

$$= \left[\mu + t\sigma^2\right]_{t=0}$$

$$= \mu$$

Variance using Cumulant Generating Function is given by

$$V(X) = \left[\frac{d^2}{dt^2}K_X(t)\right]_{t=0}$$

$$= \left[\frac{d}{dt}\left(\mu + t\sigma^2\right)\right]_{t=0}$$

$$= \left[0 + \sigma^2\right]_{t=0}$$

$$= \sigma^2$$

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